
Physics 523 HW 5

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		I use the notation in my Real Scalar Field Notes. I usually use p for the magnitude of the momentum 3-vector, $p = \mathbf{p} $, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the dot product of the momentum 4-vector: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.	
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Part I

1

1. LAGRANGIAN

The bare Lagrangian is

$$\mathcal{L}_B = (D_\mu \phi_B)^* D^\mu \phi_B - m_B^2 |\phi_B|^2 - \frac{F_B^2}{4} \quad (1)$$

where “B” stands for “Bare.”

$$F^2 = F_{\mu\nu} F^{\mu\nu}, \quad D^\mu = \partial^\mu + ie_B A_B^\mu \quad (2)$$

$$\mathcal{L}_B = [(\partial_\mu + ie_B A_{B\mu}^*) \phi_B^*](\partial^\mu + ie_B A_B^\mu) \phi_B - m_B^2 |\phi_B|^2 - \frac{F_B^2}{4} \quad (3)$$

The electromagnetic field is real $A^* = A$.

$$\mathcal{L}_B = [(\partial_\mu + ie_B A_{B\mu}) \phi_B^*](\partial^\mu + ie_B A_B^\mu) \phi_B - m_B^2 |\phi_B|^2 - \frac{F_B^2}{4} \quad (4)$$

$$\mathcal{L}_B = |\partial \phi_B|^2 - m_B^2 |\phi_B|^2 - \frac{F_B^2}{4} + ie_B A_B^\mu (\phi_B \partial_\mu \phi_B^* + \phi_B^* \partial_\mu \phi_B) - e_B^2 A_B^2 |\phi_B|^2 \quad (5)$$

$$|\partial \phi|^2 = (\partial_\mu \phi^*) \partial^\mu \phi, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (6)$$

$$\phi_B = \sqrt{Z_2} \phi, \quad A_B = \sqrt{Z_3} A, \quad m_B = \sqrt{Z_m} m, \quad e_B = Z_e e \quad (7)$$

Following Schwartz I do not put subscripts on renormalized quantities.

$$\mathcal{L}_B = Z_2 |\partial \phi|^2 - Z_2 Z_m m^2 |\phi|^2 - Z_3 \frac{F^2}{4} + \sqrt{Z_3} Z_2 Z_e ie A^\mu (\phi \partial_\mu \phi^* + \phi^* \partial_\mu \phi) - Z_2 Z_3 Z_e^2 e^2 A^2 |\phi|^2 \quad (8)$$

Like Schwartz Eq. 19.7 I define

$$Z_1 = Z_e Z_2 \sqrt{Z_3}. \quad (9)$$

$$\mathcal{L}_B = Z_2 |\partial \phi|^2 - Z_2 Z_m m^2 |\phi|^2 - Z_3 \frac{F^2}{4} + Z_1 ie A^\mu (\phi \partial_\mu \phi^* + \phi^* \partial_\mu \phi) - Z_2 Z_3 Z_e^2 e^2 A^2 |\phi|^2 \quad (10)$$

$$Z_4 = Z_2 Z_3 Z_e^2 \quad (11)$$

$$\mathcal{L}_B = Z_2 |\partial \phi|^2 - Z_2 Z_m m^2 |\phi|^2 - Z_3 \frac{F^2}{4} + Z_1 ie A^\mu J_\mu - Z_4 e^2 A^2 |\phi|^2 \quad (12)$$

$$J_\mu = \phi \partial_\mu \phi^* + \phi^* \partial_\mu \phi \quad (13)$$

$$Z_i = 1 + \delta_i, \quad Z_m = 1 + \delta_m \quad (14)$$

$$\mathcal{L}_B = (1 + \delta_2) |\partial \phi|^2 - (1 + \delta_2)(1 + \delta_m) m^2 |\phi|^2 - (1 + \delta_3) \frac{F^2}{4} + (1 + \delta_1) ie A^\mu J_\mu - (1 + \delta_4) e^2 A^2 |\phi|^2 \quad (15)$$

$$\mathcal{L}_B = (1 + \delta_2) |\partial \phi|^2 - (1 + \delta_m + \delta_2 + \delta_2 \delta_m) m^2 |\phi|^2 - (1 + \delta_3) \frac{F^2}{4} + (1 + \delta_1) ie A^\mu J_\mu - (1 + \delta_4) e^2 A^2 |\phi|^2 \quad (16)$$

The counterterms, δ_i , are infinite but formally treated as small perturbations of order e^2 . For this problem I only work to order e^2 and drop all higher order terms.

$$\mathcal{L}_B = (1 + \delta_2) |\partial \phi|^2 - (1 + \delta_m + \delta_2) m^2 |\phi|^2 - (1 + \delta_3) \frac{F^2}{4} + (1 + \delta_1) ie A^\mu J_\mu - (1 + \delta_4) e^2 A^2 |\phi|^2 \quad (17)$$

$$\mathcal{L}_B = \mathcal{L}_0 + ie A^\mu J_\mu - e^2 A^2 |\phi|^2 + \mathcal{L}_C \quad (18)$$

$$\mathcal{L}_0 = |\partial \phi|^2 - m^2 |\phi|^2 - \frac{F^2}{4} \quad (19)$$

$$\mathcal{L}_C = \delta_2 |\partial \phi|^2 - (\delta_m + \delta_2) m^2 |\phi|^2 - \delta_3 \frac{F^2}{4} + \delta_1 ie A^\mu J_\mu - \delta_4 e^2 A^2 |\phi|^2 \quad (20)$$

The propagator like counterterm vertices are order e^2 . The interaction like counterterm vertices are higher order, so I drop them.

$$\mathcal{L}_C = \delta_2 |\partial \phi|^2 - (\delta_m + \delta_2) m^2 |\phi|^2 - \delta_3 \frac{F^2}{4} \quad (21)$$

2. FEYNMAN RULES

The propagators, Fig. 1, are

$$\vec{A} \sim -\frac{i\eta_{\mu\nu}}{p^2 + i\epsilon} \quad \phi \sim \frac{i}{p^2 - m^2 + i\epsilon}. \quad (22)$$

I show the renormalized interaction vertices in Figs. 2 & 3. I show the counterterm interaction vertices in Figs. 4 & 5.

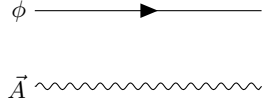
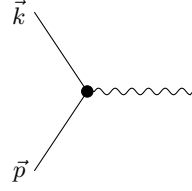


FIG. 1. Scalar QED propagators.


 FIG. 2. The 3-way renormalized vertex contributes a factor of $ie(\pm\vec{p} \pm \vec{k})$, where $-/+$ is for particles/antiparticles, see Schwartz Eqs. 9.17 - 9.20.

3. SCALAR 2-POINT FUNCTION

I want the Green's function

$$\langle \Omega | T \phi(\vec{x}) \phi(\vec{y}) | \Omega \rangle = \frac{1}{(2\pi)^4} \int d^4p e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}(\vec{p}). \quad (23)$$

In particular I want the momentum space Green's function $\mathcal{G}(\vec{p})$ to order e^2

$$\boxed{\mathcal{G}^{(2)}(\vec{p}) = \mathcal{G}^{(0)}(\vec{p}) + \Delta \mathcal{G}(\vec{p})}. \quad (24)$$

4. LEADING ORDER

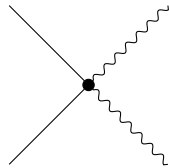
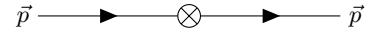
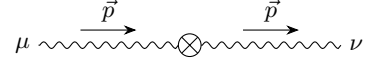
To zeroth order Green's function is given by the Feynman diagram in Fig. 6.

$$\boxed{\mathcal{G}^{(0)}(\vec{p}) = \frac{i}{p^2 - m^2 + i\epsilon}} \quad (25)$$

5. SECOND ORDER

The leading correction comes from the diagrams in Figs. 7-9.

$$\Delta \mathcal{G}(\vec{p}) = \star + \mathfrak{F} + \mathfrak{G} \quad (26)$$


 FIG. 3. The 4-way renormalized vertex contributes $2ie^2\eta_{\mu\nu}$ (Schwartz Eq. 9.23).

 FIG. 4. This scalar 2-way vertex has factor $i[p^2\delta_2 - (\delta_m + \delta_2)m^2]$, see Schwartz Eq. 19.13.

 FIG. 5. This photon 2-way vertex has factor $-i\delta_3 p^2 \eta^{\mu\nu}$, see Schwartz Eq. 19.15

$$\mathfrak{G} = \frac{i}{p^2 - m^2 + i\epsilon} i[p^2\delta_2 - (\delta_m + \delta_2)m^2] \frac{i}{p^2 - m^2 + i\epsilon} \quad (27)$$

$$\boxed{\mathfrak{G} = i \frac{m^2\delta_m + (m^2 - p^2)\delta_2}{(p^2 - m^2 + i\epsilon)^2}} \quad (28)$$

$$\mathfrak{F} = \frac{i}{p^2 - m^2 + i\epsilon} \mathbb{T} \frac{i}{p^2 - m^2 + i\epsilon} \quad (29)$$

$\mathbb{T}$ is the tadpole loop in $\mathfrak{F}$.

$$\mathfrak{F} = -\frac{\mathbb{T}}{(p^2 - m^2 + i\epsilon)^2} \quad (30)$$

$$\star = \frac{i}{p^2 - m^2 + i\epsilon} \Sigma_2 \frac{i}{p^2 - m^2 + i\epsilon} \quad (31)$$

$$\star = -\frac{\Sigma_2}{(p^2 - m^2 + i\epsilon)^2} \quad (32)$$

6. SCALAR QED COUPLING

To use DimReg I need to know the dimensions of the coupling e .

$$[\phi] = 1, \quad [x] = -1, \quad [\mathcal{L}] = 4 \quad (33)$$

$$[\mathcal{L}] = 4 \implies [F^2] = 4 \implies [F] = 2 \quad (34)$$

$$[F] = [\partial A] = \left[\frac{A}{x} \right] = [A] - [x] = [A] + 1 = 2 \implies [A] = 1 \quad (35)$$

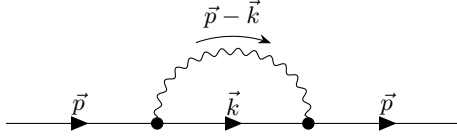
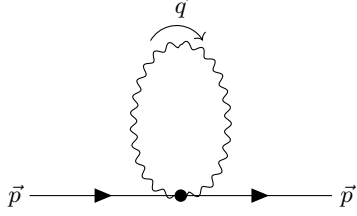
$$[\mathcal{L}] = 4 = [e^2 A^2 \phi^2] = 2[eA\phi] \implies [eA\phi] = 2 \quad (36)$$

$$[e] = 2 - [A] - [\phi] = 2 - 2 = 0 \quad (37)$$

$$\boxed{[e] = 0} \quad (38)$$

This is the same as QED, so in DimReg $e^2 \rightarrow e^2 \mu^{4-d}$ also.


 FIG. 6. $\mathcal{G}^{(0)}(\vec{p})$; the external momentum $\vec{p}$ is off-shell.


 FIG. 7. Feynman diagram $\star$, with $\vec{p}$ off-shell.

 FIG. 8. Feynman diagram $\circ$, with $\vec{p}$ off-shell.

7. TADPOLE LOOP

$$\mathbb{T} = \frac{S}{(2\pi)^4} 2ie^2 \eta_{\mu\nu} \int d^4q \frac{-i\eta^{\mu\nu}}{q^2 + i\epsilon} \quad (39)$$

S is the symmetry factor (Fig. 10).

$$S = \frac{2}{(2!)^2} = \frac{1}{2} \quad (40)$$

The denominator comes from the double A and ϕ fields.

$$\mathbb{T} = \frac{e^2}{(2\pi)^4} \eta_{\mu\nu} \eta^{\mu\nu} I \quad (41)$$

$$I = \int \frac{d^4q}{q^2 + i\epsilon} \quad (42)$$

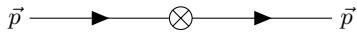
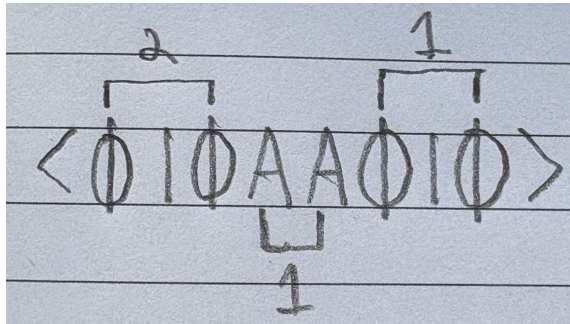

 FIG. 9. Feynman diagram $\otimes$, with $\vec{p}$ off-shell.


FIG. 10. Tadpole symmetry factor combinatorics.

I appears to diverge.

$$I \sim \int_0^\Lambda dq q \sim \Lambda^2, \quad \Lambda \rightarrow \infty \quad (43)$$

I use DimReg.

$$\mathbb{T}(d) = \frac{e^2 \mu^{4-d}}{(2\pi)^d} \eta_{\mu\nu} \eta^{\mu\nu} I(d) \quad (44)$$

$$I(d) = \int \frac{d^d q}{q^2 + i\epsilon} \quad (45)$$

$$\eta_{\mu\nu} \eta^{\mu\nu} = d \quad (46)$$

$$\mathbb{T}(d) = \frac{e^2 \mu^{4-d}}{(2\pi)^d} d I(d) \quad (47)$$

I Wick rotate I . In my Real Scalar Field Notes I derived

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int \frac{d^4 p_E}{(-p_E^2 - \Delta)^n}. \quad (48)$$

I generalize to d dimensions.

$$\int \frac{d^d p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int \frac{d^d p_E}{(-p_E^2 - \Delta)^n}. \quad (49)$$

I apply this to Eq. 45.

$$I(d) = \int \frac{d^d q}{q^2 + i\epsilon} = -i \int \frac{d^d q_E}{q_E^2} \quad (50)$$

$$q_E^2 = \sum_{n=0}^{d-1} (q^n)^2 \quad (51)$$

I take advantage of spherical symmetry using the formulas in my Real Scalar Field Notes.

$$I(d) = \int \frac{d^d q}{q^2 + i\epsilon} = -i \int \frac{d^d q_E}{q_E^2} = -i \int \frac{d^d x}{r^2} \quad (52)$$

$$d^d x = d\Omega_d r^{d-1} dr \quad (53)$$

$$I(d) = -i \int d\Omega_d \int_0^\infty dr \frac{r^{d-1}}{r^2} = -i \Omega_d \int_0^\infty dr r^{d-3} \quad (54)$$

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)} \quad (55)$$

$$I(d) = -\frac{2i\pi^{d/2}}{\Gamma(d/2)} \mathcal{A}, \quad \mathcal{A} = \int_0^\infty dr r^{d-3} \quad (56)$$

$$\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2 \quad (57)$$

$$\mathcal{A}_1 = \int_0^a dr r^{d-3}, \quad \mathcal{A}_2 = \int_a^\infty dr r^{d-3}, \quad 0 < a < \infty \quad (58)$$

$$\mathcal{A}_1 = \frac{a^{d-2}}{d-2}, \quad d > 2 \quad (59)$$

$$\mathcal{A}_2 = -\frac{a^{d-2}}{d-2}, \quad d < 2 \quad (60)$$

I analytically continue $\mathcal{A}_1$ and $\mathcal{A}_2$ to some complex d .

$$\mathcal{A} = \frac{a^{d-2}}{d-2} - \frac{a^{d-2}}{d-2} = 0 \implies I(d) = 0 \implies \mathbb{T}(d) = 0 \quad (61)$$

I analytically continue back to 4D.

$$\mathbb{T} = \mathbb{T}(4), \quad \mathbb{T}(d) = 0 \implies \boxed{\mathbb{T} = 0} \quad (62)$$

$$\mathbb{T} = 0 \implies \boxed{\mathbb{G} = 0} \quad (63)$$

8. SCALAR SELF ENERGY LOOP

$$\Sigma_2 = \frac{S}{(2\pi)^4} \int d^4k \, ie(-p^\mu - k^\mu) \frac{-i\eta_{\mu\nu}}{(\vec{p} - \vec{k})^2 + i\epsilon} ie(-p^\nu - k^\nu) \frac{i}{k^2 - m^2 + i\epsilon} \quad (64)$$

S is the symmetry factor for this diagram (Fig. 11).

$$S = \frac{4 \times 2}{(2!)^2 2!} = \frac{8}{(2!)^3} = 1 \quad (65)$$

$$\Sigma_2 = -\frac{e^2 \eta_{\mu\nu}}{(2\pi)^4} \int d^4k \frac{p^\mu + k^\mu}{(\vec{p} - \vec{k})^2 + i\epsilon} \frac{p^\nu + k^\nu}{k^2 - m^2 + i\epsilon} \quad (66)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int d^4k \frac{p^\mu + k^\mu}{(\vec{p} - \vec{k})^2 + i\epsilon} \frac{p_\mu + k_\mu}{k^2 - m^2 + i\epsilon} \quad (67)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int d^4k \frac{p^2 + k^2 + 2\vec{p} \cdot \vec{k}}{[(\vec{p} - \vec{k})^2 + i\epsilon](k^2 - m^2 + i\epsilon)} \quad (68)$$

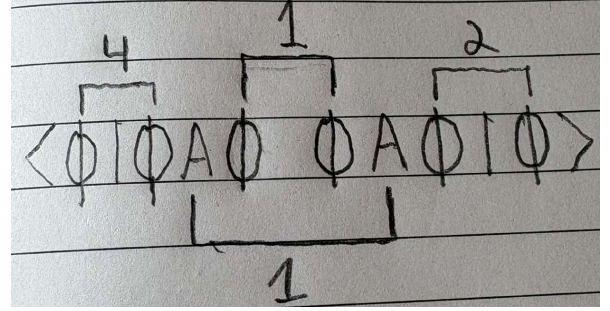


FIG. 11. Self-energy symmetry factor.

8.1. Feynman Parameter

$$A = k^2 - m^2 + i\epsilon, \quad B = (\vec{p} - \vec{k})^2 + i\epsilon \quad (69)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int d^4k \frac{p^2 + k^2 + 2\vec{p} \cdot \vec{k}}{AB} \quad (70)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (71)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{p^2 + k^2 + 2\vec{p} \cdot \vec{k}}{D^2} \quad (72)$$

$$D = k^2 - m^2 + [(\vec{p} - \vec{k})^2 - k^2 + m^2]x + i\epsilon \quad (73)$$

$$D = k^2 - m^2 + (p^2 - 2\vec{p} \cdot \vec{k} + m^2)x + i\epsilon \quad (74)$$

8.2. Complete the Square

$$D = k^2 - 2x\vec{p} \cdot \vec{k} + xp^2 + xm^2 - m^2 + i\epsilon \quad (75)$$

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (76)$$

$$a \rightarrow 1, x \rightarrow \vec{k}, b \rightarrow -2x\vec{p}, c \rightarrow xp^2 + xm^2 - m^2 + i\epsilon \quad (77)$$

$$D = (\vec{k} - x\vec{p})^2 + (1-x)(xp^2 - m^2) + i\epsilon \quad (78)$$

$$\Delta = (1-x)(m^2 - xp^2) \quad (79)$$

$$D = (\vec{k} - x\vec{p})^2 - \Delta + i\epsilon \quad (80)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{p^2 + k^2 + 2\vec{p} \cdot \vec{k}}{[(\vec{k} - x\vec{p})^2 - \Delta + i\epsilon]^2} \quad (81)$$

8.3. Shift Integrand

$$\vec{q} = \vec{k} - x\vec{p} \implies \vec{k} = \vec{q} + x\vec{p}, \quad d^4q = d^4k \quad (82)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{(x^2 + 2x + 1)p^2 + q^2 + 2(1+x)\vec{q} \cdot \vec{p}}{(q^2 - \Delta + i\epsilon)^2} \quad (83)$$

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{(1+x)^2 p^2 + q^2 + 2(1+x)\vec{q} \cdot \vec{p}}{(q^2 - \Delta + i\epsilon)^2} \quad (84)$$

The odd term cancels.

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{q^2 + (1+x)^2 p^2}{(q^2 - \Delta + i\epsilon)^2} \quad (85)$$

8.4. DimReg

$$\Sigma_2 = -\frac{e^2}{(2\pi)^4} \int_0^1 dx \mathbb{I}, \quad \mathbb{I} = \int d^4q \frac{q^2 + (1+x)^2 p^2}{(q^2 - \Delta + i\epsilon)^2} \quad (86)$$

$$\mathbb{I} = \mathbb{I}_1 + (1+x)^2 p^2 \mathbb{I}_2 \quad (87)$$

$$\mathbb{I}_1 = \int d^4q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathbb{I}_2 = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (88)$$

$$\mathbb{I}_1 \sim \int_0^\Lambda dq q \sim \Lambda^2, \quad \mathbb{I}_2 \sim \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (89)$$

$$\Sigma_2(d) = -\frac{\mu^{4-d} e^2}{(2\pi)^d} \int_0^1 dx \mathbb{I}(d) \quad (90)$$

$$\mathbb{I}(d) = \mathbb{I}_1(d) + (1+x)^2 p^2 \mathbb{I}_2(d) \quad (91)$$

$$\mathbb{I}_1(d) = \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathbb{I}_2(d) = \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2} \quad (92)$$

In my Real Scalar Field Notes I derived

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (93)$$

$$\int d^d p \frac{p^2}{(p^2 - \Delta + i\epsilon)^2} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right).$$

$$\mathbb{I}_1(d) = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right), \quad \mathbb{I}_2(d) = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (94)$$

$$\Sigma_2(d) = -\frac{\mu^{4-d} e^2}{(2\pi)^d} \int_0^1 dx [\mathbb{I}_1(d) + (1+x)^2 p^2 \mathbb{I}_2(d)] \quad (95)$$

$$\Sigma_2(d) = i \frac{\mu^{4-d} e^2}{(2\pi)^d} \pi^{d/2} \int_0^1 \frac{dx}{\Delta^{1-d/2}} \left[\frac{d}{2} \Gamma\left(\frac{2-d}{2}\right) - \frac{(1+x)^2 p^2}{\Delta} \Gamma\left(\frac{4-d}{2}\right) \right] \quad (96)$$

$$\Sigma_2(d) = i \frac{\mu^{4-d} e^2}{2^d \pi^{d/2}} \int_0^1 \frac{dx}{\Delta^{1-d/2}} \left[\frac{d}{2} \Gamma\left(1 - \frac{d}{2}\right) - \frac{(1+x)^2 p^2}{\Delta} \Gamma\left(2 - \frac{d}{2}\right) \right] \quad (97)$$

$$\Sigma_2(d) = i \frac{\mu^{4-d} e^2}{2^d \pi^{d/2}} \int_0^1 \frac{dx}{\Delta^{1-d/2}} \left[\frac{d}{2} \Gamma\left(1 - \frac{d}{2}\right) - \frac{(1+x)^2 p^2}{\Delta} \Gamma\left(1 + 1 - \frac{d}{2}\right) \right] \quad (98)$$

$$\Gamma(x+1) = x\Gamma(x) \quad (99)$$

$$\Sigma_2(d) = i \frac{\mu^{4-d} e^2}{2^d \pi^{d/2}} \Gamma\left(1 - \frac{d}{2}\right) \int_0^1 dx \Delta^{d/2-1} \left[\frac{d}{2} - \frac{(1+x)^2 p^2}{\Delta} \left(1 - \frac{d}{2}\right) \right] \quad (100)$$

8.5. Return to 4D

$$\Sigma_2 = \Sigma_2(4) = \Sigma_2(4 - \epsilon), \quad \lim \epsilon \rightarrow 0 \quad (101)$$

$$\Sigma_2 = \frac{i\mu^\epsilon e^2}{2^{5-\epsilon}\pi^{2-\epsilon/2}} \Gamma\left(\frac{\epsilon}{2} - 1\right) \int_0^1 dx \Delta^{1-\epsilon/2} \left[4 - \epsilon + \frac{(1+x)^2 p^2}{\Delta} (2-\epsilon)\right] \quad (102)$$

$$\Sigma_2 = \frac{ie^2}{32\pi^2} \Gamma\left(\frac{\epsilon}{2} - 1\right) \left(2\mu\sqrt{\frac{\pi}{\Delta}}\right)^\epsilon \int_0^1 dx \Delta \left[4 + 2\frac{p^2}{\Delta}(1+x)^2 - \left(1 + \frac{p^2}{\Delta}(1+x)^2\right)\epsilon\right] \quad (103)$$

$$h = \frac{p^2}{\Delta}(1+x)^2 \quad (104)$$

$$\Sigma_2 = \frac{ie^2}{32\pi^2} \Gamma\left(\frac{\epsilon}{2} - 1\right) \left(2\mu\sqrt{\frac{\pi}{\Delta}}\right)^\epsilon \int_0^1 dx \Delta [4 + 2h - (1+h)\epsilon] \quad (105)$$

I expand as $\epsilon \rightarrow 0$.

$$\Sigma_2 = \frac{ie^2}{16\pi^2} \int_0^1 dx \Delta \left[1 + h - (2+h)\left(\frac{2}{\epsilon} + 1 + \ln(\tilde{\mu}^2/\Delta)\right)\right] \quad (106)$$

$$\tilde{\mu}^2 = 4\pi e^{-\gamma} \mu^2 \quad (107)$$

$$\star = -\frac{\Sigma_2}{(p^2 - m^2 + i\epsilon)^2} \quad (108)$$

9. SECOND ORDER GREEN'S FUNCTION

$$\Delta\mathcal{G}(\vec{p}) = \star + \cancel{\star} + \star = \star + \star \quad (109)$$

$$\Delta\mathcal{G} = i\frac{m^2\delta_m + (m^2 - p^2)\delta_2}{(p^2 - m^2 + i\epsilon)^2} - \frac{\Sigma_2}{(p^2 - m^2 + i\epsilon)^2} \quad (110)$$

$$\Sigma_2 = i\mathbb{S} \quad (111)$$

$$\mathbb{S} = \frac{e^2}{16\pi^2} \int_0^1 dx \Delta \left[1 + h - (2+h)\left(\frac{2}{\epsilon} + 1 + \ln(\tilde{\mu}^2/\Delta)\right)\right] \quad (112)$$

$$\Delta\mathcal{G} = i\frac{m^2\delta_m + (m^2 - p^2)\delta_2 - \mathbb{S}}{(p^2 - m^2 + i\epsilon)^2} = \frac{i\mathbb{H}}{(p^2 - m^2 + i\epsilon)^2} \quad (113)$$

$$\mathbb{H} = m^2\delta_m + (m^2 - p^2)\delta_2 - \mathbb{S} \quad (114)$$

10. COUNTERTERMS

I use minimal subtraction (MS): counterterms have no finite parts.

$$\mathbb{H} = \mathbb{H}_\infty + \mathbb{H}_f, \quad (115)$$

where $\mathbb{H}_\infty$ is the divergent part of $\mathbb{H}$ and $\mathbb{H}_f$ is the finite part. MS means

$$0 = \mathbb{H}_\infty = m^2\delta_m + (m^2 - p^2)\delta_2 - \mathbb{S}_\infty, \quad (116)$$

where $\mathbb{S}_\infty$ is the divergent part of $\mathbb{S}$.

$$\mathbb{S}_\infty = -\frac{e^2}{8\pi^2\epsilon} \int_0^1 dx \Delta(2+h) \quad (117)$$

$$\mathbb{S}_\infty = -\frac{e^2}{8\pi^2\epsilon} \int_0^1 dx [2\Delta + p^2(1+x)^2] \quad (118)$$

$$\mathbb{S}_\infty = -\frac{e^2}{8\pi^2\epsilon} \int_0^1 dx [2(1-x)m^2 + (1+3x^2)p^2] \quad (119)$$

$$\mathbb{S}_\infty = -\frac{e^2}{8\pi^2\epsilon} \left[2m^2 \int_0^1 dx (1-x) + p^2 \int_0^1 dx (1+3x^2)\right] \quad (120)$$

$$\int_0^1 dx (1-x) = \frac{1}{2}, \quad \int_0^1 dx (1+3x^2) = 2 \quad (121)$$

$$\mathbb{S}_\infty = -\frac{e^2}{8\pi^2\epsilon} (m^2 + 2p^2) \quad (122)$$

I plug this into Eq. 116.

$$0 = \left(\delta_m + \delta_2 + \frac{e^2}{8\pi^2\epsilon}\right)m^2 + \left(\frac{e^2}{4\pi^2\epsilon} - \delta_2\right)p^2 \quad (123)$$

This implies

$$\frac{e^2}{4\pi^2\epsilon} - \delta_2 = \delta_m + \delta_2 + \frac{e^2}{8\pi^2\epsilon} = 0. \quad (124)$$

So

$$\frac{e^2}{4\pi^2\epsilon} = \delta_2, \quad \delta_m = -\frac{3e^2}{8\pi^2\epsilon}. \quad (125)$$

The final answer is

$$\boxed{Z_2 = 1 + \frac{e^2}{4\pi^2\epsilon}}, \quad \boxed{Z_m = 1 - \frac{3e^2}{8\pi^2\epsilon}}. \quad (126)$$

Part II

2

I follow Peskin & Schroeder (PS) Sec. 19.1 and Schwartz Sec. 18.3.

11. LAGRANGIAN

PS Eq. 19.2 gives

$$\mathcal{L} = i\bar{\psi}\not{D}\psi - \frac{F^2}{4}, \quad F^2 = F_{\alpha\beta}F^{\alpha\beta}. \quad (127)$$

In 2D $\mu = 0, 1$.

$$\mathcal{L} = i\bar{\psi}\gamma^\mu D_\mu\psi - \frac{F^2}{4}, \quad D_\mu = \partial_\mu + ieA_\mu \quad (128)$$

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - e\bar{\psi}\gamma^\mu A_\mu\psi - \frac{F^2}{4} \quad (129)$$

$$\boxed{\mathcal{L} = i\bar{\psi}\not{\partial}\psi - e\bar{\psi}\not{A}\psi - \frac{F^2}{4}} \quad (130)$$

12. DIRAC ALGEBRA

The (2×2) Dirac matrices are

$$\gamma^0 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \quad (131)$$

The metric is

$$\eta = \text{diag}(1, -1). \quad (132)$$

The Dirac anti-commutation relation is

$$\{\gamma^\alpha, \gamma^\beta\} = 2\eta^{\alpha\beta}I = 2\eta^{\alpha\beta}. \quad (133)$$

I suppress the 2D identity matrix I . I confirmed this relation with Mathematica.

13. UNITS & RENORMALIZABILITY

Using HW 4:

$$[\mathcal{L}] = d, \quad [\psi] = \frac{d-1}{2}, \quad [A] = \frac{d}{2} - 1. \quad (134)$$

For $d = 2$

$$[\mathcal{L}] = 2, \quad [\psi] = \frac{1}{2}, \quad [A] = 0. \quad (135)$$

$$[\mathcal{L}] = 2 = [e\bar{\psi}\not{A}\psi] = [e] + [\bar{\psi}\psi] + [A] \quad (136)$$

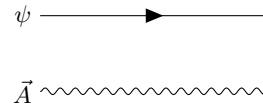


FIG. 12. 2D QED propagators.

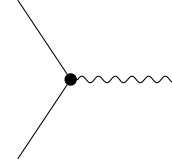


FIG. 13. The QED vertex with factor $-ie\gamma^\mu$.

$$[\psi] = [\bar{\psi}], \quad [\gamma] = 0 \quad (137)$$

$$\begin{aligned} 2 &= [e] + [\psi^2] + [A] = [e] + 2[\psi] + [\gamma A] = [e] + 2[\psi] + [\gamma] + [A] \\ &= [e] + 2[\psi] + [A] = [e] + 1 \implies \boxed{[e] = 1} \end{aligned} \quad (138)$$

This theory is super-renormalizable.

14. FEYNMAN RULES

The propagators (Fig. 12) are

$$\vec{A} \sim -\frac{i\eta_{\mu\nu}}{p^2 + i\epsilon}, \quad \psi \sim \frac{i\not{p}}{p^2 + i\epsilon}, \quad (139)$$

where the electron is massless and I use Feynman gauge for the photon. The interaction vertex is in Fig. 13.

15. PHOTON SELF-ENERGY LOOP

I evaluate the diagram in Fig. 14.

$$\begin{aligned} i\Pi_2^{\mu\nu} &= -\frac{S}{(2\pi)^2} \\ &\int d^2q \text{Tr} \left[(-ie\gamma^\mu) \frac{i(\not{q} - \not{p})}{(\vec{q} - \vec{p})^2 + i\epsilon} (-ie\gamma^\nu) \frac{i\not{q}}{q^2 + i\epsilon} \right] \end{aligned} \quad (140)$$

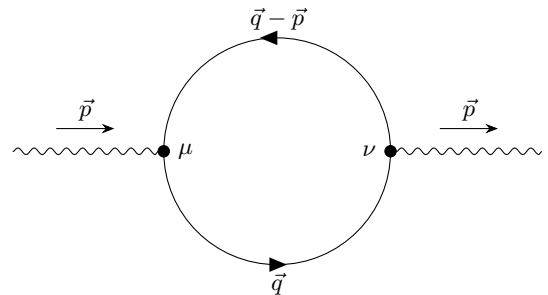


FIG. 14. Amputated photon self-energy loop: $i\Pi_2^{\mu\nu}$.

I worked out the symmetry factor in HW 2: $S = 1$.

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{(2\pi)^2} \int d^2q \frac{\text{Tr}[\gamma^\mu(\not{q} - \not{p})\gamma^\nu \not{q}]}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (141)$$

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{(2\pi)^2} \int d^2q \frac{\text{Tr}(\gamma^\mu \not{q} \gamma^\nu \not{q} - \gamma^\mu \not{p} \gamma^\nu \not{q})}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (142)$$

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{(2\pi)^2} \int d^2q \frac{\text{Tr}(\gamma^\mu \not{q} \gamma^\nu \not{q}) - \text{Tr}(\gamma^\mu \not{p} \gamma^\nu \not{q})}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (143)$$

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{(2\pi)^2} \int d^2q \frac{\text{Tr}(\gamma^\mu q_\alpha \gamma^\alpha \gamma^\nu q_\beta \gamma^\beta) - \text{Tr}(\gamma^\mu p_\alpha \gamma^\alpha \gamma^\nu q_\beta \gamma^\beta)}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (144)$$

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{(2\pi)^2} \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) \int d^2q \frac{q_\alpha q_\beta - q_\beta p_\alpha}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (145)$$

In 4D $\text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) = 4(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\alpha\nu})$. In 2D

$$\text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) = 2(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\alpha\nu}). \quad (146)$$

I confirmed this in Mathematica.

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{2\pi^2} \int d^2q \frac{N}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (147)$$

$$N = (\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\alpha\nu})(q_\alpha q_\beta - q_\beta p_\alpha) \quad (148)$$

$$N = \eta^{\mu\alpha}(q_\alpha q^\nu - q^\nu p_\alpha) - \eta^{\mu\nu}(q_\alpha q^\alpha - q^\alpha p_\alpha) + \eta^{\alpha\nu}(q_\alpha q^\mu - q^\mu p_\alpha) \quad (149)$$

$$N = q^\mu q^\nu - q^\nu p^\mu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu} + q^\nu q^\mu - q^\mu p^\nu \quad (150)$$

$$N = 2q^\nu q^\mu - q^\nu p^\mu - q^\mu p^\nu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu} \quad (151)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int d^2q \frac{2q^\nu q^\mu - q^\nu p^\mu - q^\mu p^\nu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu}}{[(\vec{q} - \vec{p})^2 + i\epsilon](q^2 + i\epsilon)} \quad (152)$$

15.1. Feynman Parameters

$$A = (\vec{q} - \vec{p})^2 + i\epsilon, \quad B = q^2 + i\epsilon \quad (153)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int d^2q \frac{2q^\nu q^\mu - q^\nu p^\mu - q^\mu p^\nu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu}}{AB} \quad (154)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (155)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2q \frac{2q^\nu q^\mu - q^\nu p^\mu - q^\mu p^\nu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu}}{D^2} \quad (156)$$

$$D = (\vec{q} - \vec{p})^2 + [q^2 - (\vec{q} - \vec{p})^2]x + i\epsilon \quad (157)$$

$$D = (\vec{q} - \vec{p})^2 + (2\vec{q} \cdot \vec{p} - p^2)x + i\epsilon \quad (158)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 + i\epsilon \quad (159)$$

15.2. Completing the Square

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a} \quad (160)$$

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow 2(x-1)\vec{p}, \quad c \rightarrow (1-x)p^2 + i\epsilon \quad (161)$$

$$D = [\vec{q} + (x-1)\vec{p}]^2 - (x-1)^2 p^2 + (1-x)p^2 + i\epsilon \quad (162)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 + (1-x)xp^2 + i\epsilon \quad (163)$$

$$\Delta = -(1-x)xp^2 \quad (164)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon \quad (165)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2q \frac{2q^\nu q^\mu - q^\nu p^\mu - q^\mu p^\nu + (\vec{q} \cdot \vec{p} - q^2)\eta^{\mu\nu}}{([\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (166)$$

15.3. Shift Integrand

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2k \frac{2k^\nu k^\mu - k^\nu p^\mu - k^\mu p^\nu + (\vec{k} \cdot \vec{p} - k^2)\eta^{\mu\nu}}{([\vec{k} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (167)$$

$$\vec{q} = \vec{k} - (1-x)\vec{p}, \quad d^2k = d^2q, \quad \vec{k} = \vec{q} + (1-x)\vec{p} \quad (168)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2q \frac{\mathcal{N}}{(q^2 - \Delta + i\epsilon)^2} \quad (169)$$

$$\begin{aligned} \mathcal{N} = & 2[q^\nu + (1-x)p^\nu][q^\mu + (1-x)p^\mu] - [q^\nu + (1-x)p^\nu]p^\mu \\ & - [q^\mu + (1-x)p^\mu]p^\nu + ([\vec{q} + (1-x)\vec{p}] \cdot \vec{p} - [\vec{q} + (1-x)\vec{p}]^2)\eta^{\mu\nu} \end{aligned} \quad (170)$$

$$\begin{aligned} \mathcal{N} = & 2q^\mu q^\nu + 2(1-x)(q^\nu p^\mu + q^\mu p^\nu) + 2(1-x)^2 p^\mu p^\nu \\ & - q^\nu p^\mu - (1-x)p^\mu p^\nu - q^\mu p^\nu - (1-x)p^\mu p^\nu + \\ & [\vec{q} \cdot \vec{p} - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - q^2 - (1-x)^2 p^2]\eta^{\mu\nu} \end{aligned} \quad (171)$$

$$\mathcal{N} = 2q^\mu q^\nu + (1-2x)(q^\nu p^\mu + q^\mu p^\nu) - 2(1-x)xp^\mu p^\nu + [(1-x)xp^2 - (1-2x)\vec{q} \cdot \vec{p} - q^2]\eta^{\mu\nu} \quad (172)$$

$$\begin{aligned} \Pi_2^{\mu\nu} = & \frac{ie^2}{2\pi^2} \int_0^1 dx \int \frac{d^2q}{(q^2 - \Delta + i\epsilon)^2} [2q^\mu q^\nu \\ & + \cancel{(1-2x)(q^\nu p^\mu + q^\mu p^\nu)} - 2(1-x)xp^\mu p^\nu \\ & + [(1-x)xp^2 - \cancel{(1-2x)\vec{q} \cdot \vec{p}} - q^2]\eta^{\mu\nu}] \end{aligned} \quad (173)$$

Odd terms integrate to 0.

$$\begin{aligned} \Pi_2^{\mu\nu} = & \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2q \\ & \frac{2q^\mu q^\nu - 2(1-x)xp^\mu p^\nu + [(1-x)xp^2 - q^2]\eta^{\mu\nu}}{(q^2 - \Delta + i\epsilon)^2} \end{aligned} \quad (174)$$

$$\begin{aligned} \Pi_2^{\mu\nu} = & \frac{ie^2}{2\pi^2} \int_0^1 dx \int d^2q \\ & \frac{2q^\mu q^\nu - \eta^{\mu\nu} q^2 - 2(1-x)xp^\mu p^\nu + \eta^{\mu\nu}(1-x)xp^2}{(q^2 - \Delta + i\epsilon)^2} \end{aligned} \quad (175)$$

$$\begin{aligned} \Pi_2^{\mu\nu} = & \frac{ie^2}{2\pi^2} \int_0^1 dx \\ & (2\mathcal{J}^{\mu\nu} - \eta^{\mu\nu}\mathbb{I} + [\eta^{\mu\nu}(1-x)xp^2 - 2(1-x)xp^\mu p^\nu]\mathfrak{J}) \end{aligned} \quad (176)$$

$$\begin{aligned} \mathcal{J}^{\mu\nu} = & \int d^2q \frac{q^\mu q^\nu}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathbb{I} = \int d^2q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2}, \\ \mathfrak{J} = & \int \frac{d^2q}{(q^2 - \Delta + i\epsilon)^2} \end{aligned} \quad (177)$$

15.4. UV Divergence Check

$$\mathfrak{J} \sim \int^\Lambda dq \frac{q}{q^4} \sim \int^\Lambda \frac{dq}{q^3} \sim \frac{1}{\Lambda^2} = 0, \quad \Lambda \rightarrow \infty \quad (178)$$

$$\mathcal{J}^{\mu\nu} \sim \mathbb{I} \sim \int_1^\Lambda dq \frac{q^3}{q^4} = \int_1^\Lambda \frac{dq}{q} = \ln \Lambda \quad (179)$$

$\mathfrak{J}$ converges, but $\mathcal{J}^{\mu\nu}$ & $\mathbb{I}$ diverge logarithmically.

15.5. Integrate $\mathfrak{J}$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx (\clubsuit^{\mu\nu} + [\eta^{\mu\nu}(1-x)xp^2 - 2(1-x)xp^\mu p^\nu]\mathfrak{J}) \quad (180)$$

$$\clubsuit^{\mu\nu} = 2\mathcal{J}^{\mu\nu} - \eta^{\mu\nu}\mathbb{I}, \quad \mathfrak{J} = \int \frac{d^2q}{(q^2 - \Delta + i\epsilon)^2} \quad (181)$$

In my Real Scalar Field Notes I derive

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (182)$$

$$\mathfrak{J} = \frac{i\pi}{\Delta} \Gamma(1) = \frac{i\pi}{\Delta}, \quad \Gamma(1) = 1 \quad (183)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \left[\clubsuit^{\mu\nu} + i\pi(1-x)x \frac{\eta^{\mu\nu}p^2 - 2p^\mu p^\nu}{\Delta} \right] \quad (184)$$

$$\Delta = -(1-x)xp^2 \quad (185)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \left[\clubsuit^{\mu\nu} - i\pi \frac{\eta^{\mu\nu}p^2 - 2p^\mu p^\nu}{p^2} \right] \quad (186)$$

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \left[\clubsuit^{\mu\nu} - i\pi \left(\eta^{\mu\nu} - \frac{2p^\mu p^\nu}{p^2} \right) \right] \quad (187)$$

15.6. DimReg

$$\begin{aligned} \clubsuit^{\mu\nu} = & 2\mathcal{J}^{\mu\nu} - \eta^{\mu\nu}\mathbb{I}, \quad \mathcal{J}^{\mu\nu} = \int d^2q \frac{q^\mu q^\nu}{(q^2 - \Delta + i\epsilon)^2}, \\ \mathbb{I} = & \int d^2q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2} \end{aligned} \quad (188)$$

$$\mathcal{J}^{\mu\nu}(d) = \int d^d q \frac{q^\mu q^\nu}{(q^2 - \Delta + i\epsilon)^2} \quad (189)$$

$$\mathbb{I}(d) = \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2} \quad (190)$$

$$\clubsuit^{\mu\nu}(d) = 2\mathcal{J}^{\mu\nu}(d) - \eta^{\mu\nu}\mathbb{I}(d) \quad (191)$$

$$\clubsuit^{\mu\nu} = \clubsuit^{\mu\nu}(2), \quad \mathcal{J}^{\mu\nu} = \mathcal{J}^{\mu\nu}(2), \quad \mathbb{I} = \mathbb{I}(2) \quad (192)$$

I use the identity (Schwartz Eq. B.51)

$$\int d^d q \frac{q^\mu q^\nu}{(q^2 - \Delta + i\epsilon)^2} = \frac{\eta^{\mu\nu}}{d} \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2}. \quad (193)$$

This identity is not kosher if either integral diverges, so I can't use it for $d = 2$.

$$\mathcal{J}^{\mu\nu}(d) = \frac{\eta^{\mu\nu}}{d} \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2} = \eta^{\mu\nu} \frac{\mathbb{I}(d)}{d} \quad (194)$$

$$\clubsuit^{\mu\nu}(d) = \eta^{\mu\nu} \left(\frac{2}{d} - 1 \right) \mathbb{I}(d) \quad (195)$$

At $d = 2$, $2/d - 1 = 0$ and $\mathbb{I}(2)$ diverges, so I need to take the limit carefully. In my Real Scalar Field Notes I derive

$$\int d^d p \frac{p^2}{(p^2 - \Delta + i\epsilon)^2} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right). \quad (196)$$

$$\mathbb{I}(d) = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) \quad (197)$$

$$\clubsuit^{\mu\nu}(d) = \frac{i}{2} \eta^{\mu\nu} \left(1 - \frac{2}{d}\right) \frac{d\pi^{d/2}}{\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) \quad (198)$$

$$\clubsuit^{\mu\nu}(d) = \frac{i\eta^{\mu\nu}}{2\Delta} (d-2)(\pi\Delta)^{d/2} \Gamma\left(1 - \frac{d}{2}\right) \quad (199)$$

$$\clubsuit^{\mu\nu} = \clubsuit^{\mu\nu}(2) = \clubsuit^{\mu\nu}(2 - \epsilon), \quad \epsilon \rightarrow 0 \quad (200)$$

$$\clubsuit^{\mu\nu} = \frac{\pi\eta^{\mu\nu}}{2i} \frac{\epsilon}{(\pi\Delta)^{\epsilon/2}} \Gamma\left(\frac{\epsilon}{2}\right) \quad (201)$$

I take $\epsilon \rightarrow 0$.

$$\clubsuit^{\mu\nu} = -i\pi\eta^{\mu\nu} \quad (202)$$

The end result is finite. This is why I didn't need counterterms: the UV divergence was an illusion.

$$\Pi_2^{\mu\nu} = \frac{ie^2}{2\pi^2} \int_0^1 dx \left[\clubsuit^{\mu\nu} - i\pi \left(\eta^{\mu\nu} - \frac{2p^\mu p^\nu}{p^2} \right) \right] \quad (203)$$

$$\Pi_2^{\mu\nu} = \frac{e^2}{\pi} \int_0^1 dx \left(\eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \quad (204)$$

$$\Pi_2^{\mu\nu} = \frac{e^2}{\pi} \left(\eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \int_0^1 dx \quad (205)$$

$$\boxed{\Pi_2^{\mu\nu} = \frac{e^2}{\pi} \left(\eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right)} \quad (206)$$

16. PHOTON PROPAGATOR

The full photon propagator $\mathcal{G}^{\mu\nu}(\vec{p})$ is defined by

$$\langle \Omega | T[A^\mu(\vec{x}) A^\nu(\vec{y})] | \Omega \rangle = \frac{i}{(2\pi)^2} \int d^2 p e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}(\vec{p}). \quad (207)$$

The sum of all amputated 1-particle irreducible (1PI) diagrams is

$$i\Pi^{\mu\nu} = \mu \bullet \text{1PI} \bullet \nu, \quad (208)$$

where the vertices (black dots) don't have photons attached to emphasize that this diagram is amputated. The gray circle represents the sum of all 1PI diagrams. From now on I omit the 1PI text. The full propagator is

$$\begin{aligned} i\mathcal{G}^{\mu\nu}(\vec{p}) = & \mu \xrightarrow{\vec{p}} \nu + \\ & \mu \xrightarrow{\vec{p}} \alpha \bullet \text{gray circle} \bullet \beta \xrightarrow{\vec{p}} \nu + \\ & \mu \xrightarrow{\vec{p}} \alpha \bullet \text{gray circle} \bullet \beta \xrightarrow{\vec{p}} \rho \bullet \text{gray circle} \bullet \sigma \xrightarrow{\vec{p}} \nu + \\ & \mu \xrightarrow{\vec{p}} \alpha \bullet \text{gray circle} \bullet \beta \xrightarrow{\vec{p}} \rho \bullet \text{gray circle} \bullet \sigma \xrightarrow{\vec{p}} \xi \bullet \text{gray circle} \bullet \gamma \xrightarrow{\vec{p}} \nu \\ & + \dots \end{aligned} \quad (209)$$

I translate into algebra (mind my careful treatment of the indices)

$$\begin{aligned} i\mathcal{G}^{\mu\nu}(\vec{p}) = & \frac{\eta^{\mu\nu}}{ip^2} + \frac{\eta^{\mu\alpha}}{ip^2} i\Pi_{\alpha\beta} \frac{\eta^{\beta\nu}}{ip^2} + \\ & \frac{\eta^{\mu\alpha}}{ip^2} i\Pi_{\alpha\beta} \frac{\eta^{\beta\rho}}{ip^2} i\Pi_{\rho\sigma} \frac{\eta^{\sigma\nu}}{ip^2} + \\ & \frac{\eta^{\mu\alpha}}{ip^2} i\Pi_{\alpha\beta} \frac{\eta^{\beta\rho}}{ip^2} i\Pi_{\rho\sigma} \frac{\eta^{\sigma\xi}}{ip^2} i\Pi_{\xi\gamma} \frac{\eta^{\gamma\nu}}{ip^2} + \dots \end{aligned} \quad (210)$$

$$\begin{aligned} i\mathcal{G}^{\mu\nu}(\vec{p}) = & -i \frac{\eta^{\mu\nu}}{p^2} - i \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\nu}}{p^2 p^2} - i \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\rho} \Pi_{\rho\sigma} \eta^{\sigma\nu}}{p^2 p^2 p^2} \\ & - i \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\rho} \Pi_{\rho\sigma} \eta^{\sigma\xi} \Pi_{\xi\gamma} \eta^{\gamma\nu}}{p^2 p^2 p^2 p^2} + \dots \end{aligned} \quad (211)$$

$$\begin{aligned} \mathcal{G}^{\mu\nu}(\vec{p}) = & -\frac{\eta^{\mu\nu}}{p^2} - \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\nu}}{p^2 p^2} - \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\rho} \Pi_{\rho\sigma} \eta^{\sigma\nu}}{p^2 p^2 p^2} \\ & - \frac{\eta^{\mu\alpha} \Pi_{\alpha\beta} \eta^{\beta\rho} \Pi_{\rho\sigma} \eta^{\sigma\xi} \Pi_{\xi\gamma} \eta^{\gamma\nu}}{p^2 p^2 p^2 p^2} + \dots \end{aligned} \quad (212)$$

$$p^{2n} = (\vec{p} \cdot \vec{p})^n \quad (213)$$

17. PHOTON MASS

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{\Pi^{\mu\nu}}{p^4} - \frac{\Pi^{\mu\rho}\Pi_{\rho}^{\nu}}{p^6} - \frac{\Pi^{\mu\rho}\Pi_{\rho\sigma}\Pi^{\sigma\nu}}{p^8} + \dots \quad (214)$$

I approximate by replacing Π with its lowest order approximation Π_2 in Eq. 206.

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{\Pi_2^{\mu\nu}}{p^4} - \frac{\Pi_2^{\mu\rho}(\Pi_2)_{\rho}^{\nu}}{p^6} - \frac{\Pi_2^{\mu\rho}(\Pi_2)_{\rho\sigma}\Pi_2^{\sigma\nu}}{p^8} + \dots \quad (215)$$

$$\Pi_2^{\mu\nu} = \frac{e^2}{\pi} P^{\mu\nu}, \quad P^{\mu\nu} = \eta^{\mu\nu} - \frac{p^{\mu}p^{\nu}}{p^2} \quad (216)$$

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{e^2 P^{\mu\nu}}{\pi p^4} - \frac{e^4 P^{\mu\rho} P_{\rho}^{\nu}}{\pi^2 p^6} - \frac{e^6 P^{\mu\rho} P_{\rho\sigma} P^{\sigma\nu}}{\pi^3 p^8} + \dots \quad (217)$$

$$\begin{aligned} P^{\mu\rho} P_{\rho}^{\nu} &= \left(\eta^{\mu\rho} - \frac{p^{\mu}p^{\rho}}{p^2} \right) \left(\eta_{\rho}^{\nu} - \frac{p_{\rho}p^{\nu}}{p^2} \right) = \\ &= \left(\eta^{\mu\rho} - \frac{p^{\mu}p^{\rho}}{p^2} \right) \left(\delta_{\rho}^{\nu} - \frac{p_{\rho}p^{\nu}}{p^2} \right) = \\ &= \eta^{\mu\rho} \delta_{\rho}^{\nu} - \eta^{\mu\rho} \frac{p_{\rho}p^{\nu}}{p^2} - \delta_{\rho}^{\nu} \frac{p^{\mu}p^{\rho}}{p^2} + \frac{p^{\mu}p^{\rho}}{p^2} \frac{p_{\rho}p^{\nu}}{p^2} = \\ &= \eta^{\mu\nu} - \frac{p^{\mu}p^{\nu}}{p^2} - \frac{p^{\mu}p^{\nu}}{p^2} + \frac{p^{\mu}p^{\nu}p^2}{p^4} = \\ &= \eta^{\mu\nu} - \frac{p^{\mu}p^{\nu}}{p^2} = P^{\mu\nu} \end{aligned} \quad (218)$$

$$P^{\mu\rho} P_{\rho\sigma} P^{\sigma\nu} = P^{\mu\rho} P_{\rho}^{\sigma} P_{\sigma}^{\nu} = P^{\mu\sigma} P_{\sigma}^{\nu} = P^{\mu\nu} \quad (219)$$

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{e^2 P^{\mu\nu}}{\pi p^4} - \frac{e^4 P^{\mu\nu}}{\pi^2 p^6} - \frac{e^6 P^{\mu\nu}}{\pi^3 p^8} + \dots \quad (220)$$

To all orders

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{e^2 P^{\mu\nu}}{\pi p^4} \sum_{n=0}^{\infty} \left(\frac{e^2}{\pi p^2} \right)^n. \quad (221)$$

This is a geometric series

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}. \quad (222)$$

$$\mathcal{G}^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{p^2} - \frac{e^2 P^{\mu\nu}}{\pi p^4} \frac{1}{1 - \frac{e^2}{\pi p^2}} \quad (223)$$

$$\boxed{p^2 \mathcal{G}^{\mu\nu}(\vec{p}) = \left(\eta^{\mu\nu} - \frac{p^{\mu}p^{\nu}}{p^2} \right) \frac{e^2}{e^2 - \pi p^2} - \eta^{\mu\nu}} \quad (224)$$

The physical photon mass (m) is the pole of the propagator (Eq. 224):

$$e^2 - \pi p^2 = 0 \implies m^2 = p^2 = \frac{e^2}{\pi}, \quad (225)$$

so

$$\boxed{m = \frac{e}{\sqrt{\pi}}}. \quad (226)$$

This has the right dimensions (Eq. 138).